Incident Escalation Practice

Introduction

The objective of this task was to simulate an incident escalation workflow in a Security Operations Center (SOC). The exercise involved creating a security incident case in TheHive, documenting the incident using a Situation Report (SITREP) in Google Docs, and designing a workflow automation process using Splunk Phantom. This task demonstrates how SOC analysts identify incidents, escalate them to higher-level analysts, and automate response processes.

Escalation Simulation using TheHive

Overview

In a SOC environment, security alerts are monitored continuously. When suspicious activity is detected, analysts create a case in an incident response platform to track and investigate the event. In this exercise, a simulated unauthorized access attempt was analyzed and escalated to Tier 2 analysts.

Step 1: Accessing TheHive Platform

TheHive platform was launched and accessed through the browser.

URL used:

```
http://localhost:9000
```

The login interface allows SOC analysts to authenticate and access the incident management dashboard.

Step 2: Logging into TheHive

The SOC analyst logged into the platform using analyst credentials.

After successful authentication, the SOC dashboard was displayed with options such as:

- Dashboard
- Alerts
- Cases
- Observables
- Tasks

Step 3: Creating a New Security Incident Case

To simulate an escalation scenario, a new case was created in the platform.

Navigation path:

```
Cases → New Case
```

The following case details were entered.

Case Details

Title:

```
Unauthorized Access on Server-Y
```

Severity:

```
High
```

TLP:

```
Amber
```

PAP:

```
Amber
```

Description:

A suspicious login attempt was detected on Server-Y originating from IP address 192.168.1.200. The activity was flagged during authentication log monitoring and corresponds to MITRE ATT&CK technique T1078 (Valid Accounts). The alert indicates possible credential misuse and requires further investigation.

After entering the details, the case was created successfully.

Step 4: Adding Observables

Observables represent indicators of compromise (IOCs) associated with the incident.

In this scenario, the suspicious IP address was added as an observable.

Navigation:

```
Observables → Add Observable
```

Observable Details:

Type:

```
IP
```

Value:

```
192.168.1.200
```

Tags:

```
suspicious
```

Adding observables helps analysts track malicious indicators and correlate them with threat intelligence sources.

Step 5: Mapping MITRE ATT&CK Technique

The activity was mapped to the MITRE ATT&CK framework to classify the attack behavior.

Technique Added:

```
T1078 – Valid Accounts
```

This technique represents attackers using stolen or compromised credentials to gain unauthorized access to systems.

Mapping incidents to MITRE techniques helps analysts understand attacker behavior and improve detection capabilities.

Step 6: Escalation to Tier 2 Analysts

The case was escalated to Tier 2 analysts for further investigation.

Escalation Summary

A high-priority alert was triggered after detecting unauthorized access attempts on Server-Y from IP address 192.168.1.200. The activity aligns with MITRE ATT&CK technique T1078 (Valid Accounts), indicating possible credential misuse. Initial investigation revealed multiple authentication attempts recorded in system logs. The source IP is considered suspicious and requires deeper analysis. Immediate actions included isolating the affected server and monitoring authentication logs for additional activity. Due to the potential security risk, the incident has been escalated to Tier 2 analysts for advanced investigation, verification of account compromise, and further incident response actions.

Situation Report (SITREP)

A Situation Report (SITREP) provides a summary of the incident for stakeholders and management.

SITREP Document

Title:

```
Situation Report (SITREP)
Unauthorized Access on Server-Y
```

Detection Time:

```
2025-08-18 13:00
```

Source IP Address:

```
192.168.1.200
```

MITRE Technique:

```
T1078 – Valid Accounts
```

Summary:

An unauthorized access attempt was detected on Server-Y during authentication log monitoring. The suspicious activity originated from IP address 192.168.1.200 and was flagged as a potential credential misuse incident. The event was mapped to MITRE ATT&CK technique T1078, which indicates the use of valid accounts for unauthorized access.

Actions Taken:

- The affected server was isolated from the network.

- Authentication logs were analyzed to identify suspicious activity.
- The incident was escalated to Tier 2 analysts.
- Monitoring was increased to detect further malicious behavior.

Workflow Automation using Splunk Phantom

Security orchestration tools automate repetitive SOC tasks. In this exercise, a simplified automation workflow was designed using Splunk Phantom.

Automation ensures faster incident response and reduces manual workload for SOC analysts.

Automated Playbook Workflow

Trigger:

```
High Severity Security Alert
```

Automation Logic:

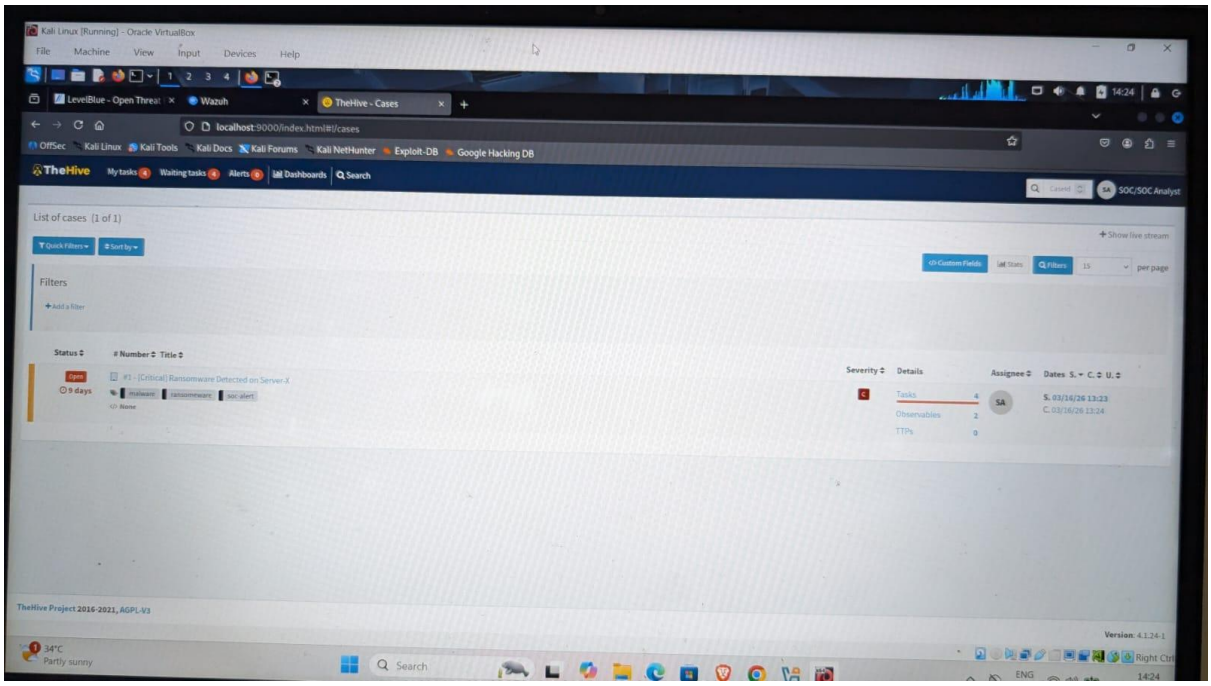
1. Receive alert from SIEM.
2. Check alert severity.
3. If severity is High:
    - Automatically assign the case to Tier 2 analysts.
    - Send notification to SOC response team.
4. Record incident activity in the case management system.

Example Playbook Logic

```
Trigger: High Severity Alert

IF severity = High
THEN
Assign Case → Tier 2 Analyst
Send Alert Notification
Log Incident Activity
END
```

This automation improves response time and ensures that critical incidents are handled by experienced analysts immediately.

Conclusion

This task demonstrated the incident escalation process in a SOC environment. A simulated unauthorized access incident was detected and investigated using TheHive platform. The suspicious activity was documented, mapped to the MITRE ATT&CK framework, and escalated to Tier 2 analysts for further investigation. A Situation Report was prepared to summarize the incident, and an automation workflow was designed using Splunk Phantom to automatically assign high-priority alerts to advanced analysts. This exercise highlights the importance of structured incident response, clear documentation, and automation in modern cybersecurity operations.